

MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 07

Level 1 — Beginner · Logical Reasoning I

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 1 — Beginner	Topic Focus	Logical Reasoning I — Patterns & Series
Questions	20	Time Allowed	30 minutes
Total Marks	20	Worksheet No.	07 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 07 opens the Logical Reasoning block — the section that makes Olympiads different from school tests. The child learns to find the rule behind number series (adding, subtracting, doubling, halving, squares), letter and day patterns, repeating groups, and odd-one-out questions. The key habit: always ask 'what is changing each time?'

Learning Objectives

- Find the rule in increasing and decreasing number series.
- Continue letter, day and shape patterns, including skip patterns.
- Use repeating groups and remainders to predict far-away positions.
- Spot the odd one out by finding what the others share.
- Recognise doubling, halving and square-number patterns.

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.

SECTION A · Multiple Choice Questions · 20 Questions

Q1. Find the next number: **3, 6, 9, 12, ____**

- (A) 13 (B) 14
(C) 15 (D) 18

Q2. Find the next number: **50, 45, 40, 35, ____**

- (A) 34 (B) 30
(C) 25 (D) 32

Q3. Find the next letter: **A, C, E, G, ____**

- (A) H (B) I
(C) J (D) K

Q4. Find the missing number: **2, 4, 8, 16, ____, 64**

- (A) 24 (B) 32
(C) 30 (D) 48

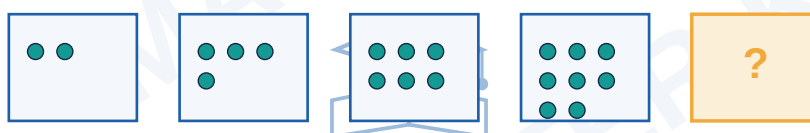
Q5. Which is the **odd one out**?

- (A) Mango (B) Banana
(C) Potato (D) Apple

Q6. Which number is the **odd one out**? **4, 8, 12, 15, 20**

- (A) 8 (B) 12
(C) 15 (D) 20

Q7. Which figure comes next? (Count the dots in each box.)



- (A) 9 dots (B) 10 dots
(C) 11 dots (D) 12 dots

Q8. Complete: **Monday, Wednesday, Friday, ____**

- (A) Saturday (B) Sunday
(C) Tuesday (D) Thursday

Q9. If the pattern is **star, moon, moon, star, moon, moon, ...** what is the **10th** shape?

- (A) Star (B) Moon
(C) Sun (D) Cannot say



Q10. Find the next number: **1, 4, 9, 16, ____**

- (A) 20 (B) 24
(C) 25 (D) 36

Q11. Which is the **odd one out**?

- (A) Triangle (B) Square
(C) Circle (D) Rectangle

Q12. Complete the pattern: **AB, CD, EF, ____**

- (A) GH (B) FG
(C) HI (D) GI

Q13. Find the missing number: **10, 20, ____, 40, 50**

- (A) 25 (B) 30
(C) 35 (D) 22

Q14. Look at the pattern: **96, 48, 24, 12, ____**. What comes next?

- (A) 10 (B) 8
(C) 6 (D) 4

Q15. Which is the **odd one out**?

- (A) Cow (B) Dog
(C) Cat (D) Crow

Q16. Find the next number: **11, 13, 15, 17, ____**

- (A) 18 (B) 19
(C) 20 (D) 21

Q17. A pattern of claps goes: 1 clap, 2 claps, 3 claps, 1 clap, 2 claps, 3 claps, ... How many claps are there in the **8th** turn?

- (A) 1 (B) 2
(C) 3 (D) 4

Q18. Complete: **Z, Y, X, W, ____**

- (A) U (B) V
(C) T (D) S

Q19. Find the missing number: **5, 11, 17, 23, ____**

- (A) 27 (B) 28
(C) 29 (D) 31

Q20. Which number should replace the question mark? **7 → 14, 9 → 18, 12 → ?**

- (A) 20 (B) 22
(C) 24 (D) 26

ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	C	B	B	B	C	C	B	B	A	C
Q	11	12	13	14	15	16	17	18	19	20
Ans	C	A	B	C	D	B	B	B	C	C

Step-by-Step Solutions

Q1. Correct answer: (C)

The pattern adds 3 each time (table of 3): $12 + 3 = 15$.

Q2. Correct answer: (B)

The pattern subtracts 5 each time: $35 - 5 = 30$.

Q3. Correct answer: (B)

The pattern skips one letter each time: A, (b), C, (d), E, (f), G, (h), I.

Q4. Correct answer: (B)

Each number is doubled: $16 \times 2 = 32$, and $32 \times 2 = 64$. ✓

Q5. Correct answer: (C)

Mango, banana and apple are fruits. **Potato** is a vegetable, so it is the odd one out.

Q6. Correct answer: (C)

4, 8, 12 and 20 are all in the table of 4 (multiples of 4). **15** is not, so it is the odd one out.

Q7. Correct answer: (B)

The dots increase by 2 each time: 2, 4, 6, 8, so the next box has **10 dots**.

Q8. Correct answer: (B)

The pattern skips one day each time: Monday, (Tue), Wednesday, (Thu), Friday, (Sat), **Sunday**.

Q9. Correct answer: (A)

The group (star, moon, moon) repeats every 3. $10 \div 3 = 3$ groups remainder 1, so the 10th shape is the FIRST of a new group — a **star**.

Q10. Correct answer: (C)

These are square numbers: 1×1 , 2×2 , 3×3 , 4×4 , so next is $5 \times 5 = 25$. (The gaps also grow: +3, +5, +7, +9.)

Q11. Correct answer: (C)

Triangle, square and rectangle have straight sides and corners. A **circle** has neither, so it is the odd one out.

Q12. Correct answer: (A)

Letters move in pairs through the alphabet: AB, CD, EF, **GH**.

Q13. Correct answer: (B)

Counting by tens: 10, 20, **30**, 40, 50.

Q14. Correct answer: (C)

Each number is HALVED: $12 \div 2 = 6$.

**Q15. Correct answer: (D)**

Cow, dog and cat are four-legged land animals. A **crow** is a bird that flies, so it is the odd one out.

Q16. Correct answer: (B)

These are odd numbers increasing by 2: $17 + 2 = 19$.

Q17. Correct answer: (B)

The group (1, 2, 3) repeats every 3 turns. $8 \div 3 = 2$ groups remainder 2, so the 8th turn is the **SECOND** of a group — **2 claps**.

Q18. Correct answer: (B)

The letters go **BACKWARDS** through the alphabet: after W comes **V**.

Q19. Correct answer: (C)

The pattern adds 6 each time: $23 + 6 = 29$.

Q20. Correct answer: (C)

Each number is doubled: $7 \rightarrow 14$, $9 \rightarrow 18$, so $12 \rightarrow 12 \times 2 = 24$.

